



HARRIS ACADEMY PECKHAM
GCSE Geography | OCR B | Year 11

PHYSICAL FIELDWORK: RESULTS

Does longshore drift occur at Whitstable?



Lesson 3: Presentation and analysis

Name	
Class	
Data collected	Monday 28 September 2026, Whitstable

In silence. You have five minutes.

(1) Which direction did our float travel, and what was the mean time over 10 m? [2] Last Lesson	(2) Which side of a groyne collects more sediment, the updrift side or the downdrift side? [1] Last Term
(3) Define swash and backwash. [2] Last Year	(4) We measured on one morning, on a rising spring tide. Give one way that could have affected our beach profile results. [3] Stretch & Challenge

1. What you found at Whitstable

Stage 3

Your other results. The two tables you need for the graphs are printed again next to each graph, so you never have to turn back.

The float survey

Attempt	Time (s)	Direction	Distance (m)
1	48	West	10
2	61	West	10
3	55	West	10
MEAN	54.7	West	—

The sediment

Measured	WEST of groyne	EAST of groyne
Mean pebble size	37.5 mm	52.3 mm
Mean roundness	4.1 (rounder)	3.0 (more angular)

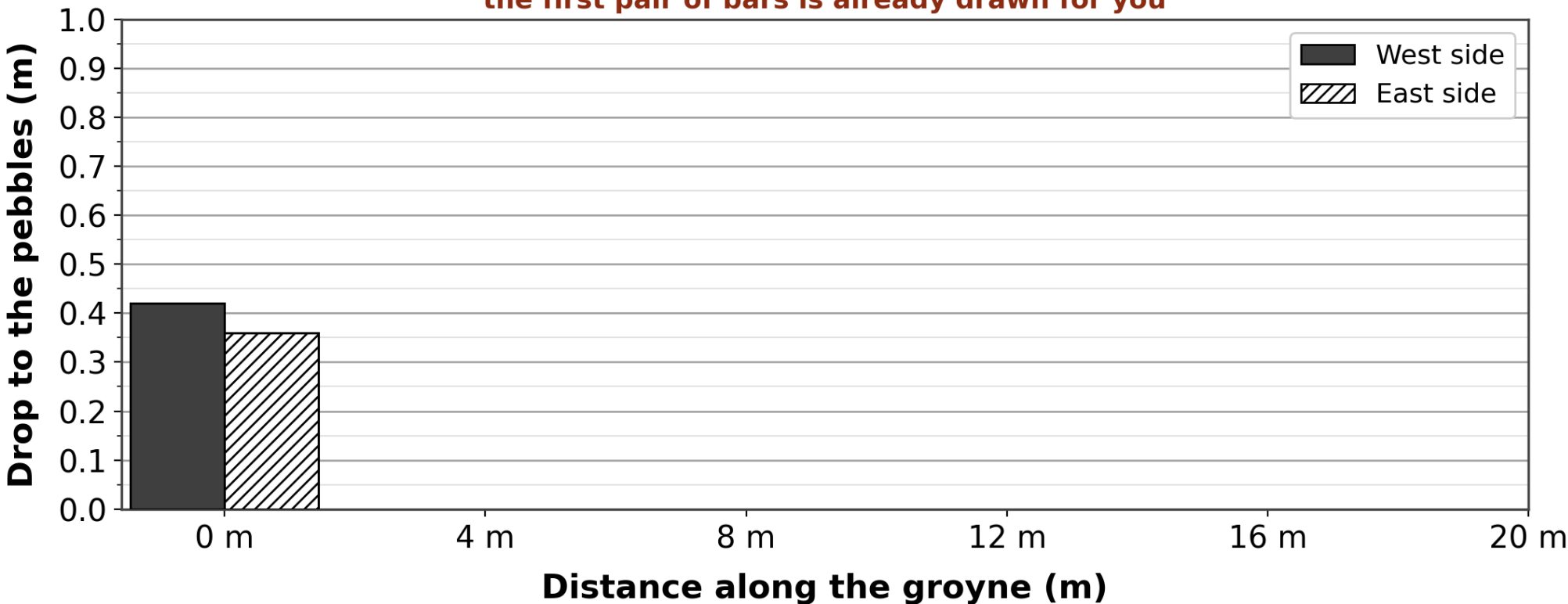
What you saw, as well as what you measured

- Shingle along the whole beach. No sand above the low water mark.
- At the water's edge the groyne sat almost level with the shingle on both sides.
- Near the sea wall, vegetation was growing between the pebbles on both sides.

Axes and the first pair are done. Draw the other five pairs: west solid, east shaded. Greyed figures below are the ones already on the graph.

Groyne 3: drop on each side

the first pair of bars is already drawn for you



Distance	0 m	4 m	8 m	12 m	16 m	20 m
West (m)	0.42	0.68	0.91	0.95	0.74	0.51
East (m)	0.36	0.49	0.58	0.59	0.52	0.42

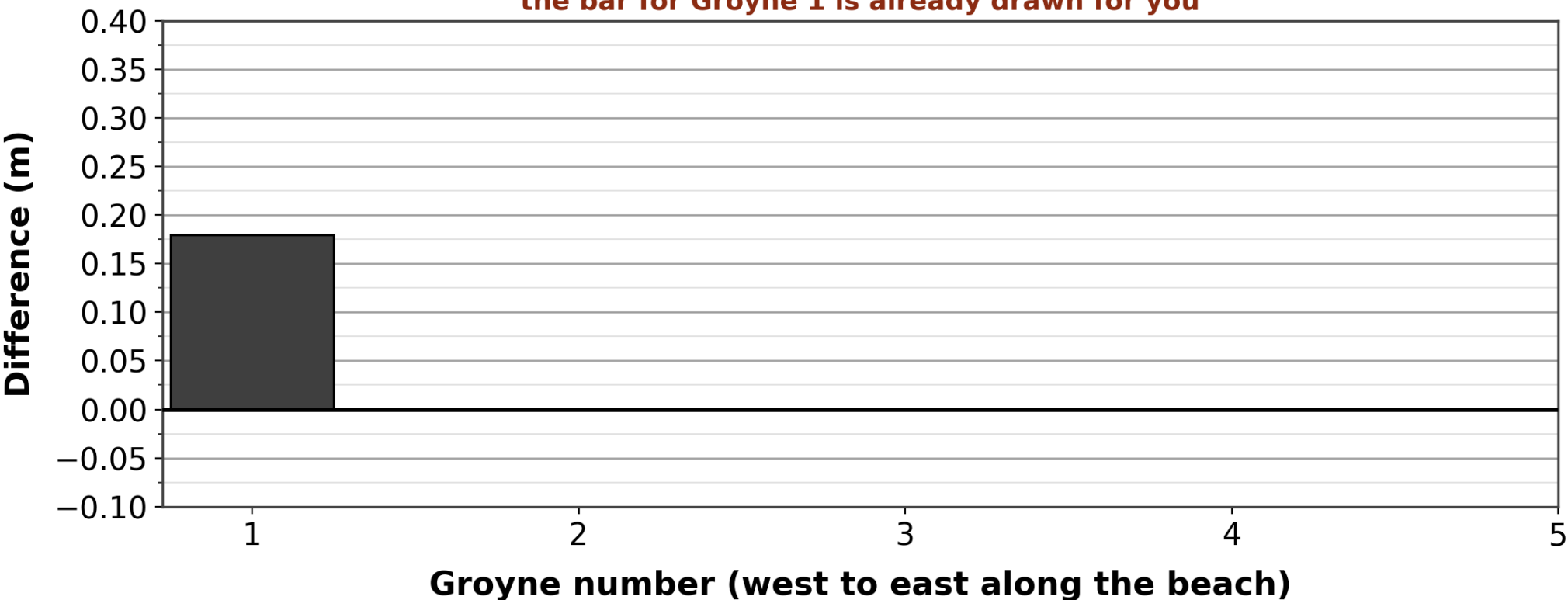
Graph 2: all five groynes

Stage 3

Groyne 1 is drawn. Draw groynes 2, 3, 4 and 5, then circle the one that does not fit.

All five groynes: west drop minus east drop

the bar for Groyne 1 is already drawn for you



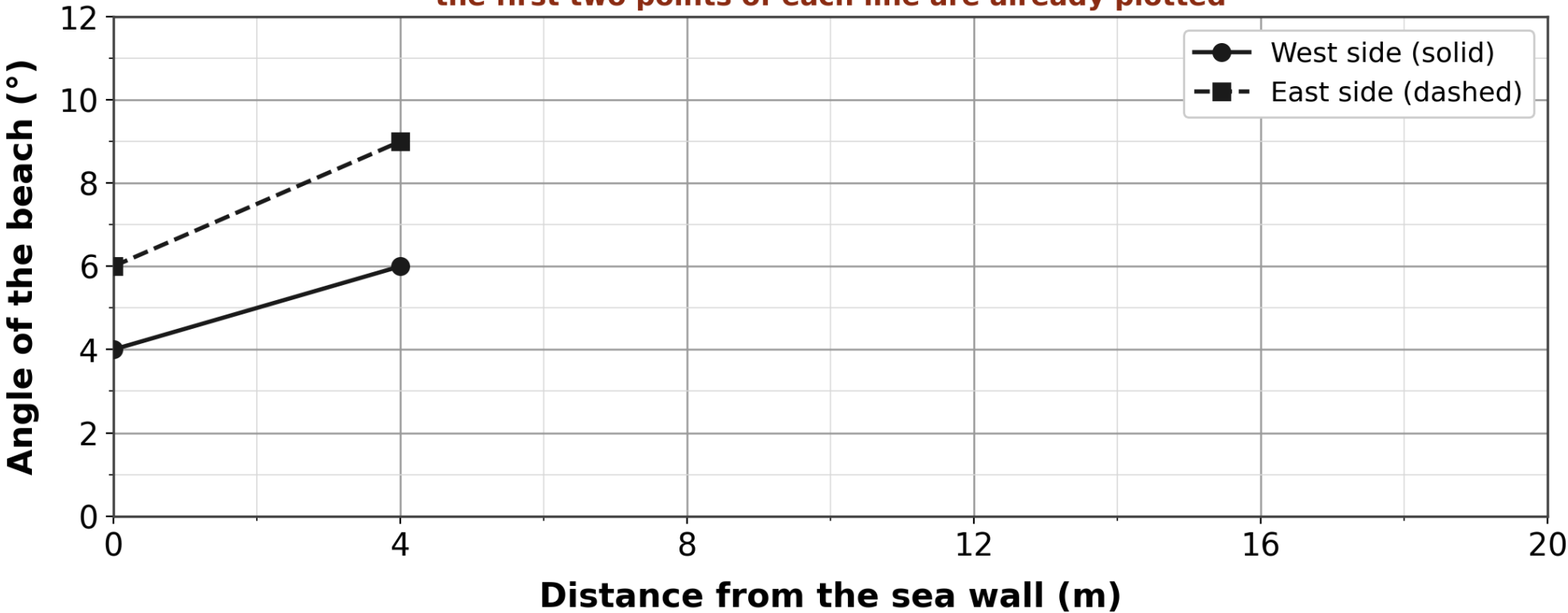
Groyne	1	2	3	4	5
Difference (m)	0.18	0.19	0.21	0.24	0.05

Which groyne did you circle? Groyne _____

First two points of each line are plotted. Plot the other four on each, then join them up.

Beach profile either side of Groyne 3

the first two points of each line are already plotted



Distance from sea wall	0 m	4 m	8 m	12 m	16 m	20 m
West side (°)	4	6	7	5	4	2
East side (°)	6	9	11	8	5	3

5. Describe what your graphs show

Stage 4: Data analysis

Fill every gap. All the numbers you need are in the box.

Number bank

west east 0.06 0.21 0.36 0.05 12 water's edge
Groyne 5

1. The drop to the pebbles was greater on the _____ side at every distance.
2. The biggest difference between the two sides was _____ m.
3. That biggest difference was at _____ m along the groyne.
4. The difference was smallest at the _____ and at the sea wall.
5. The mean difference along the whole groyne was _____ m.
6. The groyne that does not fit the class pattern is _____ .
7. Its difference was only _____ m.

Now say it in one sentence, using two of those numbers.

6. Explain why

Stage 4: Data analysis

Word bank

north east angle swash backwash gravity east west
trapped vegetation waves

1. The strongest winds at Whitstable come from the _____ .
2. This means waves hit the beach at an _____ .
3. Shingle is carried diagonally up the beach by the _____ .
4. It is then pulled straight back down by the _____ , because of _____ .
5. Repeating this moves shingle along the beach towards the _____ .
6. When the shingle reaches a groyne it gets _____ on the _____ side.
7. That is why the drop to the pebbles is smaller on that side.

Harder: why is the difference so small at BOTH ends of the groyne?

Tick the TWO reasons you think are right.

- ☐ At the water's edge the waves rework the shingle on both sides every tide.
- ☐ The groyne is broken at both ends.
- ☐ Near the sea wall there was vegetation, so waves rarely reach that far up.
- ☐ Nobody measured those parts properly.

7. Homework part 1: your conclusion

Stage 5

Write the five sentences below straight out, filling the gaps. That is your conclusion.

	Write this sentence
1	Longshore drift _____ (does / does not) occur at Whitstable, and it moves shingle from east to _____ .
2	I know this because the drop to the pebbles was smaller on the east side of the groyne, with a mean difference of _____ m.
3	A second piece of evidence is the sediment: pebbles west of the groyne were smaller, at _____ mm, and rounder than those on the east.
4	My first hypothesis was _____ (accepted / rejected) and my second hypothesis was _____ (accepted / rejected).
5	However, Groyne _____ did not fit the pattern, so I am confident but not certain.

8. Homework part 2: your evaluation

Stage 6

To what extent was your primary data collection successful? [8 marks + 3 SPaG]

Write TWO paragraphs, one for each method below. Each paragraph uses the same four sentence starters.

Method 1	Method 2
The groyne profile	The float survey

	Sentence	Start it like this
1	What I did	"We measured the drop from the top of the groyne every 4 m on both sides..."
2	What went wrong	"This was not fully accurate because..."
3	What I would change	"Next time I would... which would make the results more..."
4	What that does to my answer	"This means my conclusion is still / is not reliable because..."

Two methods done properly beats five in a list. Name Whitstable, name your groyne, use your numbers.

To what extent was your primary data collection successful? [8 marks + 3 SPaG]

Two paragraphs: one on the groyne profile, one on the float survey. Four sentences in each.

[illegible]